Kubernetes Service: ExternalName

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When you need to expose a pod, you should create a service and you should map the service to the pod.

However, you may map a service to an external domain as well.

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Prerequisite:

I have a namespace: web

I have 2 pods in the `web` namespace: **hello-nginx, hello-apache**

I have a namespace: cache

I have a pod in the `cache` namespace: **redis-cache**

You can map your service to an external resource:

```
apiVersion: v1
kind: Service
metadata:
  name: external-db-access
  namespace: web
spec:
  type: ExternalName
  externalName: google.com
```

Make sure that your service has created:

```
kubectl describe svc/external-db-access -n web
```

Connect to the pod and check the DNS address of the service `external-db-access`:

```
➜  ~ kubectl exec -it hello-nginx -n web -- /bin/bash
```

```
root@hello-nginx:/# curl external-db-access
<!DOCTYPE html>
...
```

So, when I try to access to the service `external-db-access` in the namespace `web`, I will be forwarded to `google.com` through DNS.

Kubernetes will create a CNAME record on `external-db-access` service with the value `google.com`

Can I access `external-db-access` in another namespace?

Yes and no.

I have another pod in the `cache` namespace.

```
➜ ~ kubectl exec -it redis-cache -n cache — /bin/bash
root@redis-cache:/data# curl external-db-access
curl: (6) Could not resolve host: external-db-access
```

I can't contact this domain directly since `external-db-access` is in another namespace.

I need to specify the full FQDN of the external-db-access service in the `cache` namespace:

```
➜ ~ kubectl exec -it redis-cache -n cache — /bin/bash
root@redis-cache:/data# curl external-db-access.web.svc.cluster.local
<!DOCTYPE html>
...
```

Is there another use case for ExternalName?

If you want to access a service in another namespace, you may need ExternalName.

I will delete the service `external-db-access` in the `web` access:

```
➜ ~ kubectl delete svc/external-db-access -n web
service “external-db-access” deleted
```

I will expose the pod `hello-nginx` :

```
apiVersion: v1
kind: Service
metadata:
  labels:
    app: hello-nginx-expose
  name: hello-nginx-expose
  namespace: web
spec:
  ports:
  - name: 8181-80
    port: 8181
    protocol: TCP
    targetPort: 80
  selector:
    run: hello-nginx
  type: NodePort
```

I will try to access the pod `hello-nginx` from the pod `hello-apache` :

```
➜ ~ kubectl exec -it hello-apache -n web — /bin/bash
root@hello-apache:~# curl hello-nginx-expose:8181
<!DOCTYPE html>
...
```

Since the pod `hello-nginx` and the pod `hello-apache` in the same namespace, I can access it.

I can’t access it in the pod `redis-cache` . Because it is in another namespace:

```
➜ ~ kubectl exec -it redis-cache -n cache — /bin/bash
root@redis-cache:/data# curl hello-nginx-expose:8181
```

```
curl: (6) Could not resolve host: hello-nginx-expose
```

I can access via full FQDN though:

```
➜  ~ kubectl exec -it redis-cache -n cache -- /bin/bash
root@redis-cache:/data# curl hello-nginx-expose.web.svc.cluster.local:8181
<!DOCTYPE html>
...
```

If you don't want to access the service `hello-nginx-expose` via full FQDN, you may want to create a service in the `cache` namespace:

```
apiVersion: v1
kind: Service
metadata:
  name: hello-nginx-expose-reference
  namespace: cache
spec:
  type: ExternalName
  externalName: hello-nginx-expose.web.svc.cluster.local
```

`hello-nginx-expose-reference` will be an alias for `hello-nginx-expose.web.svc.cluster.local` :

```
➜ ~ kubectl exec -it redis-cache -n cache — /bin/bash
root@redis-cache:~# curl hello-nginx-expose-reference:8181
<!DOCTYPE html>
...
```

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